Candidates' Name ……...……

Signature: ……...……

Random N o.					Personal No.		

(Do not write your school / Center name or Number anywhere)

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PHYSICS
JULY / AUG, 2026

2 HOURS: 30 MINUTES

KAMTEC EXAMINATIONS BOARD

Uganda Certificate of Education

PHYSICS

Paper 1

TIME: 2 HOURS: 30 Minutes INSTRUCTIONS

TO CANDIDATES:

This paper has **two** *sections;* **A** *and* **B.** *It has* ***seven*** *examination items.*

Section **A** *has* **three** *compulsory items.*

Section **B** *has* **two** *parts;* **I** *and* **II**.

Answer **one** *item from* **each** *part.*

Answer **five** *items in all.*

Any additional item(s) answered will not be scored.

All answers **must** *be written in the booklets provided.*

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SECTION A

Item 1

In a certain village where quarrying is done by using dynamites, an officer from the Environmental authority was sent to investigate. He wanted to view inside the perimeter wall of the company that does mining without being seen. He had a plane mirror periscope with him but he was advised to use a prism periscope. He however did not know what it looks like and what makes it better. He agreed to use it anyway.

Just as he thought of setting up his periscope, he heard a loud explosion followed by the shaking of the wall. He wondered how the explosion was linked to the shaking of the wall and if the sound from the explosion was harmful to his ears.

Just as he rode back on a motorcycle, he was surprised to see a diminished image of himself in the mirror. He was curious to know why it appeared like that and identify other purposes for which such a mirror can be used.

Hint;

Velocity of sound in air = 320 ms^{-1}

Wavelength of waves from the explosion = 2 cm

Use your knowledge of Physics to;

a. Help the officer understand how the prism periscope can be used to solve his problem, and why it should be preferred to the one he had.

b. Determine if the sound he heard was harmful to his ears.

c. Explain how the sound from the explosion reached the wall.

d. Help the officer understand how the mirror is able to form images like the one it did.

Item 2

A certain radioactive isotope is meant to be used for the generation of electricity by bombarding it with a neutron in a certain nuclear facility according to the equation below.

$$^{50}_{22}Rv + {}^{1}_{0}n \longrightarrow {}^{a}_{b}Xn + 2{}^{4}_{2}He + \text{Energy}$$

The workers are however having very varying opinions about the process. Some are worried that it might result into the production of a radioactive isotope of $^{42}_{18}Pl$ which is dangerous but are not sure of the dangers. They were also advised to wear lead jackets all the time but were not sure why. Some of the workers are curious about the precautions they need to carry out to ensure their safety and identify uses of the radioactive nuclide.

Use your knowledge of Physics;

a. Determine if the product formed is an isotope of the mentioned element.

b. Explain why there was a need to dress as recommended.

c. Identify safety measures that should be taken and help the workers understand how nuclides similar to $^{42}_{18}$Pl can be put to use.

Item 3

A student visited a certain town far away from his school, and was surprised to see that it was night time despite his watch (set to his home time) showing it was day time. On observing the shape of the moon, its appearance kept on changing periodically which surprised him since he thought the moon was spherical.

During day time, the temperatures were very high. The student was surprised that whenever power was on, and temperatures inside the house exceeded 36 ^{o}C, an air conditioner automatically switched itself on. This surprised the learner as he did not know how it works.

The student wondered when the rains would come because his electricity bill very high. He wondered if it was possible to tell when it would rain again.

Task

Use your knowledge of Physics to;

a. Help the student understand why there was a difference in time between the two places.
b. Explain why the moon appeared the way it did.

c. Help the student understand how the air conditioner is able to work automatically.

d. Briefly describe how it is possible to tell when the rains would come.

SECTION B

*(Attempt only **one** item in this part).*

Item 4

The water level in an underground tank has fallen below 11 m. This has slowed down farm operations due to their lift pump not being able to work successfully. Upon consultation, the owner of the farm was told to replace the current pump with a force pump. The farm owner demanded the farm manager to explain to him the reason behind the advice and explain to her how it works. During the set-up of the new pump, a goat accidentally fell into the well. A

veterinary doctor who was 1 km away was called said the goat needed medical attention in not more than 20 minutes after being removed from the tank.

As soon as it was removed, a cyclist set off from rest and accelerated at a rate of 1.5 ms^{-2} for 10 seconds. The cyclist maintained this speed for a certain time and then decelerated at a rate of 2.5 ms^{-2} before reaching the doctor's home. The cyclist spent 12 minutes at the doctor's home and rode back, following the same motion in reverse.

The cyclist was surprised to see the veterinary doctor put the medicine in a shiny container with ice in it. The doctor also insisted they both wear helmets but he did not explain why. The farm manager and cyclist approached you demanding explanations.

Hint: ***The medicine would become ineffective if its temperature reached room temperature.***

Task

Use your knowledge of Physics to:

a. Help the farm manager respond to the farm owner.
b. Determine if the veterinary doctor arrived in time.
c. Help the cyclist understand why the medicine was kept in such a container and why the doctor insisted on them wearing helmets.

Item 5

A mother went to shop with her family to a supermarket. After loading a trolley with items, she pushed it at a speed of 10 ms^{-1} but there was an irritating squeaking noise in the hinges of its tyres despite the tyres and floor being smooth.

As she was walking along a slope, she fainted and lost control of the trolley. It accelerated downwards at a rate of 2 ms^{-2} for 10s towards a glass window that would break if it knocked it with a velocity higher than 26 ms^{-1}.

A person who saw it said, structural modifications should be made to include a sponge on the front part of the trolley to reduce its impact on the glasses in case it knocked them. He also emphasized the need for windows to be opened as the place seemed very hot inside.

During a meeting about the incident, one of the workers requested that the tap that supplies water has very low pressure despite it having no mechanical problem. He requested that its overhead tank is raised through 4 m above its current height. No one in the meeting could understand what he meant.

Task

Use your knowledge of Physics to; (a*)*

Determine if the glass broke.

(b) Suggest ways of reducing the noise made by the trolley.

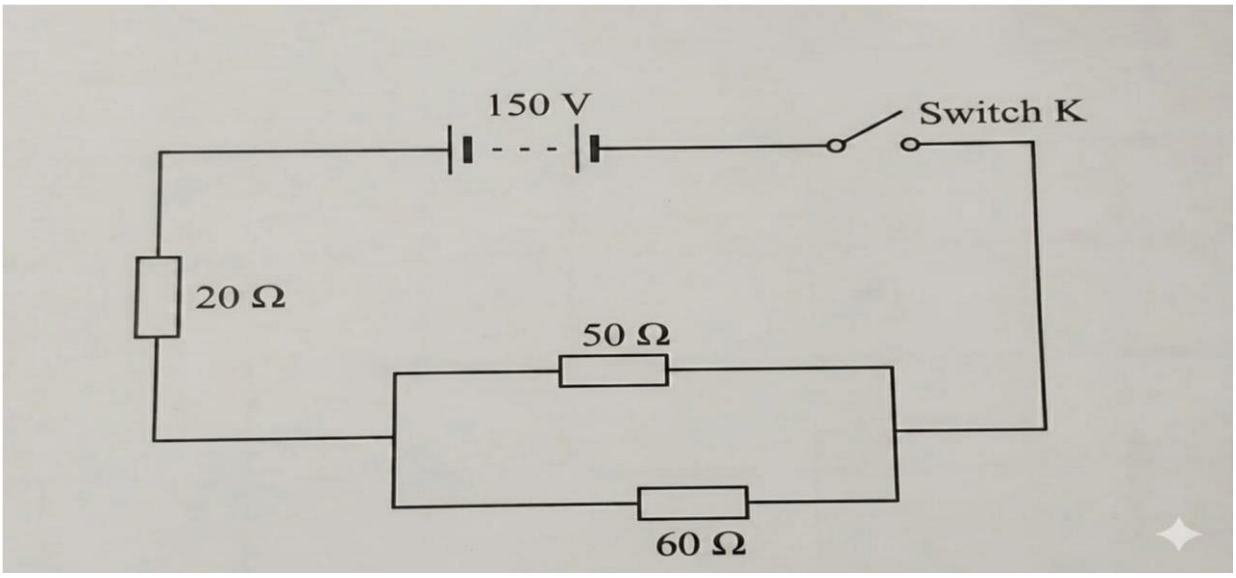
(c) Explain if the modifications made by the person would help in solving the problems in the supermarket.

PART II

*(Attempt only **one** item in this part).*

Item 6

After attempts by a homeowner to use appliances were in vain, he decided to consult an electrician. On analyzing the circuit, the electrician discovered the circuit was like the one below.



He advised the homeowner to close switch K but he argued in response. The homeowner also expressed his worry that;

- The battery was not strong enough to power the 50 Ω appliance that requires 1.50 A to work normally.
- In case the current in the circuit exceeded 4A, the 20 Ω appliance would get damaged according to his usual manual. He needed to know how to protect it from damage.
- An electric kettle in the house shocked whoever touches it when in power. After it was taken to his neighbor's house, he was surprised there were no shocks when it was touched. He wanted to know if it had a problem.

Task

Use your knowledge of Physics to;

(a) *Settle the argument between the electrician and the homeowner.*

(b) *Help the home owner determine if battery was strong enough.*

(c) *Advise the home owner on how to protect his appliance.*

(d) *Help the homeowner determine if the kettle had a problem.*

Item 7

A business man that intends to open a laundry mart in a room connected to a 240V AC mains supply recently bought a washing machine that uses direct current to power a motor. He was told that for the machine to work, the alternating current needs to change to direct current.

Just before the laundry mart had opened, he told a friend of his that the only money available to pay their weekly electrical bill is sh.90000 and requested him for ways of increasing the efficiency of the machine in order to do more work in less time. His friend also did not know how to do so since he didn't even know how the machine works.

Hint; The power rating of the washing machine was 240V, 3000 W.

Cost of electricity = Ush.800 per kWh

The machine will be used for 6 hours each day.

Task

Use your knowledge of Physics to;

(a) Help your friend understand how alternating current will be used to power the machine.

(b) Determine if the money available will be enough for the electricity bill.

(c) Explain to the two friends how the machine works and identify ways in which its efficiency can be improved.

END